

Day 1

Objective for the Day

To understand the internship project and know what work I will be doing during the internship.

List of Activities Done Today

Today was the first day of my internship. I attended the introduction session and got to know about the project that I would be working on, which is **Smart Classroom Behavior Detection using YOLOv8**. I understood the project objectives and discussed the basic workflow with my mentor. I also got a brief idea about Computer Vision and how AI can be used to detect student behaviors in a classroom.

Critical Observations

I realized that before starting any project, it is important to understand the problem clearly and have a proper plan for implementation.

What I Learned Today

I learned about the project requirements and got an introduction to Computer Vision and object detection.

What I Would Have Done Better

I could have spent some extra time reading more about the project so that I would understand it in greater detail.

Plan for Tomorrow

Tomorrow I will learn more about Computer Vision and study the YOLOv8 model.

Day 2

Objective for the Day

To learn the basics of Computer Vision and understand how it is used in real-world applications.

List of Activities Done Today

Today I spent time learning about Computer Vision and how computers can understand images and videos. I also read about object detection and came across different models used for it. Since my project is based on YOLOv8, I started learning about its features and why it is widely used for real-time object detection.

Critical Observations

I noticed that Computer Vision is used in many areas like security, healthcare, self-driving cars, and smart classrooms. It is interesting to see how AI can recognize objects automatically.

What I Learned Today

I learned the basics of Computer Vision, object detection, and the importance of YOLOv8 in AI applications.

What I Would Have Done Better

I could have watched a few more videos to understand the concepts in more detail.

Plan for Tomorrow

Search for a suitable dataset and understand the different classroom behavior classes.

Day 3

Objective for the Day

To collect a suitable dataset for the project.

List of Activities Done Today

Today I searched for classroom behavior datasets on different platforms. After comparing a few options, I selected the Smart Classroom Behavior Dataset from Kaggle. I downloaded the dataset and checked the images and behavior classes to understand what kind of data I would be working with.

Critical Observations

I realized that choosing the right dataset is very important because the model's performance depends on the quality of the data.

What I Learned Today

I learned how to find datasets and understand their structure before using them for training.

What I Would Have Done Better

I could have compared one or two more datasets before finalizing the best one.

Plan for Tomorrow

Prepare the dataset using Roboflow.

Day 4

Objective for the Day

To prepare the dataset for training.

List of Activities Done Today

Today I uploaded the dataset to Roboflow. I checked the annotations and organized the images properly. I also split the dataset into training, validation, and testing sets and exported it in YOLOv8 format so that it could be used for model training.

Critical Observations

I understood that proper dataset preparation helps in achieving better model performance and reduces errors during training.

What I Learned Today

I learned how to use Roboflow for dataset management and exporting datasets in YOLO format.

What I Would Have Done Better

I could have explored more preprocessing options available in Roboflow.

Plan for Tomorrow

Set up Google Colab and prepare the environment for training.

Day 5

Objective for the Day

To prepare the development environment.

List of Activities Done Today

Today I worked on setting up Google Colab for my project. I installed the required Python libraries and connected the dataset. I also checked whether all the folders and files were arranged correctly before starting the implementation.

Critical Observations

Even a small mistake in the file paths can cause errors, so checking everything carefully is important.

What I Learned Today

I learned how to use Google Colab and why it is useful for deep learning projects.

What I Would Have Done Better

I could have organized my project files in a better way from the beginning.

Plan for Tomorrow

Install YOLOv8 and verify that everything is working correctly.

Day 6

Objective for the Day

To install YOLOv8 and prepare the project for training.

List of Activities Done Today

Today I installed the Ultralytics YOLOv8 package and imported the required libraries. After that, I loaded the dataset and checked whether everything was configured correctly. I also understood the basic commands used for training the model.

Critical Observations

Setting up the environment correctly before training saves a lot of time later.

What I Learned Today

I learned how to install and configure YOLOv8 using Python and Google Colab.

What I Would Have Done Better

I could have explored a few more commands provided by the Ultralytics library.

Plan for Tomorrow

Start training the YOLOv8 model.

Day 7

Objective for the Day

To begin training the model.

List of Activities Done Today

Today I started training the YOLOv8n model using the prepared dataset. I configured the training settings and monitored the initial progress. It was interesting to see how the model started learning from the dataset.

Critical Observations

The training process takes time, and using a GPU helps complete it much faster.

What I Learned Today

I learned how the training process works and why pretrained models are useful.

What I Would Have Done Better

I could have read more about different YOLO model versions before training.

Plan for Tomorrow

Continue training the model and observe its performance.

Day 8

Objective for the Day

To monitor the training progress.

List of Activities Done Today

Today I continued the model training and monitored the loss values and evaluation metrics. I observed how the model performance improved after every few epochs. I also saved the training results for future reference.

Critical Observations

As the training continued, I noticed that the model gradually became more accurate.

What I Learned Today

I learned about Precision, Recall, mAP, and how they are used to evaluate a model.

What I Would Have Done Better

I could have understood the evaluation graphs in more detail.

Plan for Tomorrow

Complete the training and analyze the results.

Day 9

Objective for the Day

To complete the training process.

List of Activities Done Today

Today the model training was completed successfully. I reviewed the training graphs and checked the final performance metrics. I also saved the best model so that it could be used for testing on images and videos.

Critical Observations

The final results showed that the model was able to detect classroom behaviors with good accuracy.

What I Learned Today

I learned how to interpret the training results and identify the best-performing model.

What I Would Have Done Better

I could have tried different training settings to compare the results.

Plan for Tomorrow

Test the trained model on classroom images.

Day 10

Objective for the Day

To test the trained model using classroom images.

List of Activities Done Today

Today I tested the trained model on classroom images. The model detected different student behaviors and displayed them with bounding boxes and confidence scores. I reviewed the prediction results and saved the output images for documentation.

Critical Observations

The model performed well on most images, but I noticed that the confidence score varied depending on the image quality and student positions.

What I Learned Today

I learned how to perform image inference using YOLOv8 and analyze the prediction results.

What I Would Have Done Better

I could have tested the model on more images with different classroom conditions.

Plan for Tomorrow

Test the model on classroom videos and observe its real-time detection performance.

Day 11

Objective for the Day

To test the trained model using classroom videos.

List of Activities Done Today

Today I tested the trained model on classroom videos. I uploaded a sample video and observed how the model detected different student behaviors in each frame. I checked whether the detections were correct and saved the output video for future reference.

Critical Observations

The model was able to detect multiple students at the same time. However, in some frames, the confidence score was slightly lower due to lighting and camera angle.

What I Learned Today

I learned how YOLOv8 performs real-time object detection on videos and how each frame is processed individually.

What I Would Have Done Better

I could have tested the model using videos recorded from different classroom environments.

Plan for Tomorrow

Review the video detection results and identify areas where the model can be improved.

Day 12

Objective for the Day

To analyze the prediction results from video testing.

List of Activities Done Today

Today I carefully reviewed the output video generated by the model. I observed the detected behaviors, confidence scores, and bounding boxes. I also noted a few cases where the model missed detections or gave lower confidence.

Critical Observations

The detection accuracy depends on image clarity, lighting conditions, and the position of students inside the classroom.

What I Learned Today

I learned that testing is important to identify the strengths and limitations of a trained model.

What I Would Have Done Better

I could have tested more videos with different classroom conditions.

Plan for Tomorrow

Implement behavior counting and analyze the detected activities.

Day 13

Objective for the Day

To implement behavior counting in the detection system.

List of Activities Done Today

Today I modified the project to count the detected student behaviors. I checked whether the counts were updating correctly while processing the video. After a few corrections, the counting feature worked properly.

Critical Observations

Behavior counting provides a better understanding of classroom activities instead of showing only detections.

What I Learned Today

I learned how object detection results can be converted into useful classroom analytics.

What I Would Have Done Better

I could have displayed the behavior counts in a more organized format.

Plan for Tomorrow

Review the complete workflow and test the updated model again.

Day 14

Objective for the Day

To verify the updated project after implementing behavior counting.

List of Activities Done Today

Today I tested the updated model on both images and videos. I checked whether the detections and behavior counts were displayed correctly. I also verified that the model was working without errors.

Critical Observations

The updated system worked as expected and produced better output by displaying behavior counts along with detections.

What I Learned Today

I learned the importance of testing after making changes to the project.

What I Would Have Done Better

I could have tested the system using a larger number of sample videos.

Plan for Tomorrow

Organize the project files and save the important outputs.

Day 15

Objective for the Day

To organize the project files and outputs.

List of Activities Done Today

Today I arranged all the project files, including the trained model, dataset, prediction images, and output videos. I also created separate folders to keep everything organized for future use.

Critical Observations

Maintaining project files properly makes documentation and future modifications much easier.

What I Learned Today

I learned the importance of organizing project files and maintaining backups.

What I Would Have Done Better

I could have organized the folders from the beginning of the project.

Plan for Tomorrow

Review the complete implementation process and collect screenshots for documentation.

Day 16

Objective for the Day

To collect project outputs for documentation.

List of Activities Done Today

Today I selected the best prediction images, training graphs, and output videos. I also collected screenshots from Google Colab and Roboflow that would be useful while preparing the internship report.

Critical Observations

Screenshots help explain the project more clearly in reports and presentations.

What I Learned Today

I learned the importance of documenting every stage of the project with proper evidence.

What I Would Have Done Better

I could have captured screenshots regularly instead of collecting them at the end.

Plan for Tomorrow

Review the project from beginning to end before starting documentation.

Day 17

Objective for the Day

To review the overall implementation of the project.

List of Activities Done Today

Today I reviewed the entire project workflow from dataset collection to final detection. I checked whether all the objectives had been completed and verified that the outputs were correct. I also made a few final corrections before starting the report.

Critical Observations

Reviewing the project helped me understand the complete workflow and identify small areas for improvement.

What I Learned Today

I gained confidence in explaining the implementation process and understood the project better.

What I Would Have Done Better

I could have maintained more detailed notes throughout the implementation.

Plan for Tomorrow

Begin documenting the project and preparing the internship report.

Day 18

Objective for the Day

To prepare an overview of the project.

List of Activities Done Today

Today I started documenting the overall project. I wrote the project title, purpose, problem statement, proposed solution, and details about the dataset used. This helped me summarize the project before preparing the final report.

Critical Observations

Writing the project overview helped me understand the complete project in a simple and structured way.

What I Learned Today

I learned how to present technical information in an easy-to-understand format.

What I Would Have Done Better

I could have started documenting the project a little earlier.

Plan for Tomorrow

Continue documenting the project methodology and implementation.

Day 19

Objective for the Day

To document the project methodology.

List of Activities Done Today

Today I prepared the methodology section by writing the project workflow step by step. I described dataset collection, preprocessing, model training, testing, and behavior detection. I also arranged the workflow diagram for better understanding.

Critical Observations

A clear workflow makes it easier for others to understand the project implementation.

What I Learned Today

I learned how to organize technical content in a logical sequence.

What I Would Have Done Better

I could have included more diagrams while documenting the methodology.

Plan for Tomorrow

Prepare the results and conclusion sections of the project.

Day 20

Objective for the Day

To complete the project documentation.

List of Activities Done Today

Today I worked on the results and conclusion of the project. I selected the final prediction images, reviewed the performance of the model, and summarized the overall outcome. I also checked the report for any corrections before moving on to the presentation.

Critical Observations

Preparing the documentation made me realize how much I had learned throughout the internship and how each stage contributed to the final project.

What I Learned Today

I learned the importance of technical documentation and presenting project results clearly.

What I Would Have Done Better

I could have documented my work regularly instead of waiting until the final stage.

Plan for Tomorrow

Start preparing the PowerPoint presentation for the internship review.

Day 21

Objective for the Day

To start preparing the internship report.

List of Activities Done Today

Today I started writing the internship report. I first went through the university template and understood the required format. I organized the project details and started writing the introduction and objectives. I also collected some screenshots that I wanted to include in the report.

Critical Observations

Following the correct report format is important because it makes the document neat and easy to understand.

What I Learned Today

I learned how to organize technical information into a proper report format.

What I Would Have Done Better

I could have started writing the report while working on the project instead of waiting until the end.

Plan for Tomorrow

Continue writing the methodology and implementation sections.

Day 22

Objective for the Day

To continue preparing the internship report.

List of Activities Done Today

Today I completed the methodology section of the report. I explained each step of the project, from dataset collection to model training and testing. I also added workflow diagrams and arranged the content properly.

Critical Observations

Writing each step in sequence makes it easier for anyone reading the report to understand the project.

What I Learned Today

I learned how to present technical work in a simple and organized manner.

What I Would Have Done Better

I could have included more diagrams to explain the methodology.

Plan for Tomorrow

Complete the results and conclusion sections.

Day 23

Objective for the Day

To complete the remaining sections of the report.

List of Activities Done Today

Today I wrote the results and conclusion of the project. I selected suitable prediction images and included the final outputs in the report. After completing the content, I checked the formatting and corrected a few mistakes.

Critical Observations

Good documentation is as important as the implementation because it explains the work done during the internship.

What I Learned Today

I learned how to summarize project results and present them clearly.

What I Would Have Done Better

I could have reviewed the report more carefully before finalizing it.

Plan for Tomorrow

Start preparing the PowerPoint presentation.

Day 24

Objective for the Day

To create the internship presentation.

List of Activities Done Today

Today I started preparing the PowerPoint presentation. I created the slides according to the university template and added the project title, company details, objectives, and methodology. I also selected suitable images for each slide.

Critical Observations

A presentation should contain only important points while the explanation can be given during the review.

What I Learned Today

I learned how to prepare a professional technical presentation.

What I Would Have Done Better

I could have planned the slide design before adding the content.

Plan for Tomorrow

Complete the remaining presentation slides.

Day 25

Objective for the Day

To complete the PowerPoint presentation.

List of Activities Done Today

Today I completed the remaining slides, including implementation, results, and conclusion. I added screenshots, workflow diagrams, and project outputs. I also checked whether the slide sequence was correct.

Critical Observations

Using relevant images makes the presentation more attractive and easier to explain.

What I Learned Today

I learned how to summarize a technical project within a limited number of slides.

What I Would Have Done Better

I could have improved the design of a few slides to make them look more professional.

Plan for Tomorrow

Review the report and presentation.

Day 26

Objective for the Day

To review all project documents.

List of Activities Done Today

Today I carefully reviewed the internship report and presentation. I corrected grammatical mistakes, checked image quality, and verified the project details. I also made sure that all the figures and headings were placed correctly.

Critical Observations

Even small corrections improve the overall quality of the report and presentation.

What I Learned Today

I learned the importance of reviewing work before submission.

What I Would Have Done Better

I could have proofread the report more than once.

Plan for Tomorrow

Prepare for the project review and presentation.

Day 27

Objective for the Day

To prepare for the internship review.

List of Activities Done Today

Today I practiced explaining the project from the beginning to the end. I revised the methodology, implementation, and results so that I could explain them confidently. I also reviewed the important technical terms related to the project.

Critical Observations

Understanding the project is more important than memorizing the presentation.

What I Learned Today

I learned that confidence comes from understanding the project thoroughly.

What I Would Have Done Better

I could have practiced speaking more fluently.

Plan for Tomorrow

Revise the project once again and prepare for possible questions.

Day 28

Objective for the Day

To revise the complete project.

List of Activities Done Today

Today I revised the complete project once again. I reviewed the workflow, outputs, and important concepts related to YOLOv8 and Computer Vision. I also checked all the project files to make sure everything was ready.

Critical Observations

Revising the project helped me remember each step more clearly and increased my confidence.

What I Learned Today

I learned the importance of being well prepared before a technical presentation.

What I Would Have Done Better

I could have practiced answering technical questions in more detail.

Plan for Tomorrow

Prepare all the required documents for submission.

Day 29

Objective for the Day

To organize all the internship documents.

List of Activities Done Today

Today I arranged all the project files, internship report, presentation, screenshots, and supporting documents. I checked that everything was complete and ready for submission. I also maintained my notebook to keep track of the work completed during the internship.

Critical Observations

Keeping all the documents organized makes the submission process easier and avoids missing important files.

What I Learned Today

I learned the importance of proper planning and documentation throughout a project.

What I Would Have Done Better

I could have maintained my project notes more regularly from the beginning.

Plan for Tomorrow

Complete the internship successfully and reflect on my learning experience.

Day 30

Objective for the Day

To complete the internship and summarize my overall learning.

List of Activities Done Today

Today I completed all the internship work and reviewed the entire project one final time. I organized the report, presentation, notebook, and project files. Looking back, I realized how much I had learned during these 30 days, from understanding Computer Vision to training a YOLOv8 model and documenting the complete project.

Critical Observations

This internship gave me practical exposure to AI and helped me understand how classroom concepts are applied in real projects.

What I Learned Today

I improved my knowledge of Computer Vision, YOLOv8, Python, Google Colab, Roboflow, technical documentation, and presentation skills. I also became more confident in handling a complete AI project independently.

What I Would Have Done Better

If I had more time, I would improve the model further by testing it on more classroom datasets and adding more advanced features.

Plan for the Future

Continue learning advanced Computer Vision and Deep Learning concepts and apply them in future academic and industry projects.